Current Makes Fields

ChatGPT edition | Investigation 3

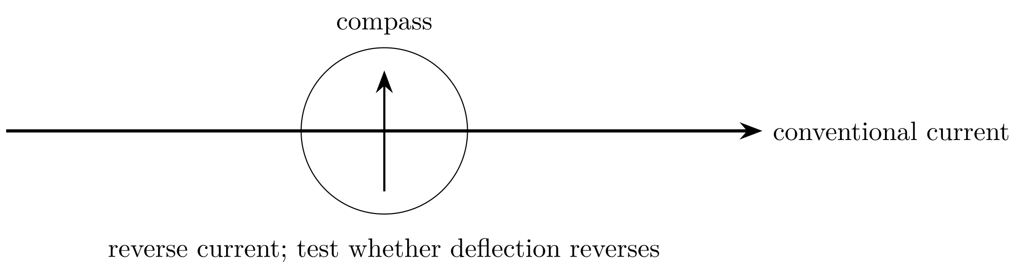
Build a compass map, then run one fair electromagnet test

SHORT CIRCUITS HEAT QUICKLY

Use two AA cells, a pushbutton, and the specified current-limiting resistor. Power only while taking a reading, never more than five seconds. A coil is a long conductor and can heat. Disconnect immediately for warmth, odor, damaged enamel, or a swollen/leaking cell. **STOP: no outlets, mains leads, wall adapters, large battery packs, large capacitors, ignition coils, or exposed high voltage.**

Part 1 — A current has direction

Lay a straight insulated wire north–south above a compass. Put a 47 Ω resistor and momentary pushbutton in series with two AA cells. Predict the needle direction, press briefly, record, then disconnect and reverse the battery holder. Repeat with the wire below the compass.



Evidence record	
Prediction	____________________
Changed variable	____________________
Measured result	____________________
Claim + because	____________________

Part 2 — Choose one variable, not three

Wind 20 neat turns of enamelled wire on an iron nail; scrape only the two end contacts. Connect the same 47 Ω resistor, pushbutton and two-cell holder. Count identical steel paper clips lifted after a two-second press.

Trial design	1	2	3	Mean
Baseline: 20 turns				
Your change: ____________				

Choose *one*: number of turns (keep cells, resistor, nail and press time); or core/no core (keep every other variable). Three repeats reveal variation. Do not remove the resistor to “make it stronger.”

Part 3 — Energy bookkeeping

The battery’s chemical store transfers energy electrically. The magnetic field mediates force; lifted clips gain gravitational energy, while wire and resistor gain thermal energy. A motor uses current and magnetic interaction to produce motion. It does not turn electricity directly into “force,” and it does not create energy.

Claim quality

Weak: “More turns is stronger.” Strong: “With the same cells, resistor, nail and two-second pulse, 40 turns lifted a mean of ______ clips versus ______ for 20 turns; the repeats ranged ______–______.”

Troubleshoot with tests

Symptom	Test one cause at a time
No compass deflection	Check closed path; rotate straight wire north–south; move it closer without touching compass.
No clips lifted	Verify enamel is removed only at contacts; test continuity with power disconnected; use a ferromagnetic steel nail.
Battery or wire warms	Disconnect immediately. Adult quarantines the cell; inspect for a direct short or damaged insulation.

Evidence: DOE S06. Measurement units: NIST S04. Safety boundary: OSHA S01–S03. See the provider-neutral electricity source register.